

**ZIGUINCHOR, Senegal**WMO: **616950**Lat: **12.556N**Lon: **16.282W**Elev: **23**StdP: **101.05**Time Zone: **0.00 (E00)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 17.0       | 17.9 | 2.5                           | 4.5 | 26.8 | 4.1 | 5.1 | 26.8 | 7.2                   | 25.9 | 6.1 | 27.0 | 1.5                   | 0    | 0.309 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 6             | 10.4                      | 39.1            | 22.2 | 38.0 | 22.4 | 36.9 | 22.8 | 28.7                | 33.2 | 28.2 | 32.5 | 27.8 | 31.9 | 2.7                     | 40   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.5                            | 23.5 | 31.2 | 27.1 | 22.9 | 30.7 | 26.8 | 22.5 | 30.3 | 93.2          | 33.5 | 90.9 | 32.4 | 88.8 | 32.2 | 35.3              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 5.8               | 5.0  | 4.4 | 14.0                       | 41.7 | 2.2                | 0.7 | 12.4                                               | 42.2 | 11.1       | 42.6 | 9.9        | 43.0 | 8.3        | 43.5 |
|                   |      |     | 9.9                        | 30.2 | 1.3                | 1.5 | 9.0                                                | 31.3 | 8.2        | 32.1 | 7.5        | 33.0 | 6.6        | 34.0 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 28.1   | 25.6  | 27.4  | 28.8  | 29.1  | 29.4  | 29.6  | 28.2  | 27.7  | 27.9  | 28.8  | 28.1  | 26.0 |
|                                                                                  | DBStd       | 1.85   | 1.62  | 1.54  | 1.50  | 1.52  | 1.36  | 1.24  | 1.30  | 1.18  | 1.27  | 1.26  | 1.30  | 1.52 |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDD10.0     | 6589   | 483   | 488   | 582   | 573   | 602   | 589   | 564   | 548   | 538   | 582   | 544   | 495  |
|                                                                                  | CDD18.3     | 3547   | 224   | 254   | 324   | 323   | 344   | 339   | 306   | 290   | 288   | 324   | 294   | 237  |
|                                                                                  | CDH23.3     | 35424  | 2045  | 2562  | 3327  | 3183  | 3506  | 3851  | 3092  | 2662  | 2764  | 3510  | 2896  | 2026 |
|                                                                                  | CDH26.7     | 17289  | 1049  | 1495  | 2040  | 1907  | 1926  | 1856  | 1173  | 887   | 1033  | 1554  | 1393  | 977  |
| Wind                                                                             | WSAvg       | 2.0    | 2.1   | 2.2   | 2.2   | 2.2   | 2.3   | 2.2   | 2.0   | 2.0   | 1.8   | 1.6   | 1.5   | 2.0  |
| Precipitation                                                                    | PrecAvg     | 1293   | 0     | 0     | 0     | 0     | 4     | 100   | 309   | 429   | 339   | 105   | 3     | 1    |
|                                                                                  | PrecMax     | 2010   | 7     | 4     | 1     | 0     | 27    | 265   | 549   | 748   | 711   | 361   | 40    | 14   |
|                                                                                  | PrecMin     | 698    | 0     | 0     | 0     | 0     | 0     | 21    | 44    | 126   | 148   | 7     | 0     | 0    |
|                                                                                  | PrecStd     | 279    | 1     | 1     | 0     | 0     | 6     | 60    | 104   | 159   | 108   | 76    | 7     | 2    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 36.2  | 39.0  | 40.7  | 40.8  | 39.7  | 37.2  | 34.9  | 33.9  | 34.6  | 35.9  | 36.1  | 36.2 |
|                                                                                  |             | MCWB   | 20.0  | 21.1  | 21.8  | 22.1  | 23.5  | 26.2  | 27.2  | 28.0  | 28.3  | 28.2  | 25.3  | 21.9 |
|                                                                                  | 2%          | DB     | 35.0  | 37.3  | 39.2  | 39.0  | 37.9  | 36.2  | 33.8  | 32.9  | 33.3  | 34.9  | 35.0  | 34.7 |
|                                                                                  |             | MCWB   | 19.7  | 20.4  | 21.4  | 22.7  | 24.3  | 26.2  | 27.2  | 27.6  | 27.8  | 28.0  | 25.1  | 21.5 |
|                                                                                  | 5%          | DB     | 33.8  | 36.2  | 37.9  | 37.5  | 36.2  | 35.1  | 32.9  | 32.0  | 32.8  | 34.0  | 34.1  | 33.2 |
|                                                                                  |             | MCWB   | 19.5  | 20.2  | 21.4  | 22.9  | 24.6  | 26.2  | 27.1  | 27.2  | 27.6  | 27.7  | 24.9  | 21.4 |
|                                                                                  | 10%         | DB     | 32.5  | 34.8  | 36.2  | 35.9  | 35.1  | 34.1  | 31.9  | 31.0  | 31.8  | 33.0  | 33.1  | 32.0 |
|                                                                                  |             | MCWB   | 19.1  | 20.0  | 21.5  | 23.1  | 24.7  | 26.1  | 26.7  | 27.0  | 27.2  | 27.4  | 24.6  | 20.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 23.9  | 25.2  | 25.9  | 26.5  | 27.5  | 28.2  | 28.4  | 28.7  | 29.1  | 29.6  | 28.8  | 26.2 |
|                                                                                  |             | MCDB   | 31.7  | 33.4  | 35.7  | 35.7  | 35.3  | 33.7  | 32.8  | 32.6  | 33.6  | 34.0  | 33.4  | 31.4 |
|                                                                                  | 2%          | WB     | 22.2  | 23.7  | 24.7  | 25.5  | 26.6  | 27.5  | 27.8  | 28.1  | 28.4  | 28.8  | 27.7  | 24.7 |
|                                                                                  |             | MCDB   | 29.8  | 32.1  | 34.2  | 34.7  | 34.1  | 33.2  | 32.1  | 31.7  | 32.4  | 33.3  | 32.3  | 29.5 |
|                                                                                  | 5%          | WB     | 21.2  | 22.7  | 23.8  | 24.7  | 25.9  | 27.0  | 27.4  | 27.5  | 27.8  | 28.2  | 27.0  | 23.6 |
|                                                                                  |             | MCDB   | 29.0  | 31.0  | 33.0  | 33.7  | 33.0  | 32.9  | 31.5  | 31.0  | 31.5  | 32.5  | 31.4  | 28.1 |
|                                                                                  | 10%         | WB     | 20.2  | 21.6  | 23.0  | 24.0  | 25.3  | 26.6  | 27.0  | 27.1  | 27.4  | 27.8  | 26.3  | 22.6 |
|                                                                                  |             | MCDB   | 28.1  | 29.9  | 31.8  | 32.4  | 32.1  | 32.1  | 30.9  | 30.3  | 30.8  | 31.7  | 30.0  | 27.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 14.6  | 15.9  | 16.8  | 16.2  | 13.8  | 10.4  | 7.9   | 7.2   | 8.0   | 9.2   | 11.3  | 13.1 |
|                                                                                  |             | MCDBR  | 16.4  | 17.6  | 18.6  | 17.6  | 15.2  | 11.6  | 9.1   | 8.5   | 9.2   | 10.1  | 12.3  | 15.1 |
|                                                                                  |             | MCWBR  | 6.6   | 6.4   | 6.1   | 5.5   | 4.7   | 3.6   | 3.6   | 3.9   | 4.3   | 4.3   | 4.5   | 5.7  |
|                                                                                  | 5% WB       | MCDBR  | 15.2  | 16.1  | 17.0  | 16.1  | 13.3  | 10.4  | 8.5   | 8.0   | 8.7   | 9.7   | 10.5  | 13.1 |
|                                                                                  |             | MCWBR  | 6.2   | 6.5   | 6.3   | 5.7   | 4.6   | 3.7   | 3.8   | 3.9   | 4.3   | 4.5   | 4.4   | 5.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.531  | 0.578 | 0.668 | 0.657 | 0.678 | 0.673 | 0.601 | 0.544 | 0.537 | 0.580 | 0.538 | 0.507 |      |
|                                                                                  | taud        | 1.913  | 1.812 | 1.641 | 1.713 | 1.685 | 1.762 | 1.932 | 2.095 | 2.093 | 1.954 | 1.995 | 2.024 |      |
|                                                                                  | Ebn at Noon | 771    | 753   | 697   | 702   | 675   | 669   | 723   | 776   | 786   | 743   | 761   | 781   |      |
|                                                                                  | Edh at Noon | 190    | 218   | 264   | 244   | 246   | 225   | 191   | 164   | 165   | 187   | 174   | 167   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 5.42   | 6.04  | 6.74  | 7.00  | 6.74  | 5.89  | 4.98  | 4.75  | 5.03  | 5.42  | 5.31  | 5.14  |      |
|                                                                                  | RadStd      | 0.27   | 0.25  | 0.22  | 0.16  | 0.26  | 0.24  | 0.26  | 0.29  | 0.25  | 0.22  | 0.23  | 0.24  |      |

**Historical Trends**

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page